MILITRY SPECIFICATION

GRAIN, IGNITER, PROPELLANT

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of propellant assembly containing pyrotechnic materials, carbon graphite spacer, and rubber pads. The propellant assembly is referenced herein as the "grain".

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-763	Steel Bars, Shapes, and Forgings - Corrosion Resisting

Military

MIL-I-82437	Igniter

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should by addressed to: Commanding Officer, Naval Ordnance Station, (524), Indian Head, MD 20640 by using the self-addressed Standardization/ Documentation Improvement Proposal (DD Form 1426) appearing at the end of this documents or by letter.

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 inch X 48 inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
AN 565	Setscrew-Hexagon and Fluted Socket, Headless
MS 16625	Ring, Retaining, Internal, Basic (Tapered Section Type)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Ordnance Systems Command

DL 2541708	Shipping (Packing) Condition (For Eight Propellant Assemblies)
DL 2541709	Propellant Can Assembly
Dwg 3209534	Propellant Assembly
Dwg 2571090	Combustion Chamber

PUBLICATIONS

Naval Ordnance Systems Command (Code Ident. 10001)

WD 30369	Preparation and Calibration, Proofing Configuration, Torpedo Mk 46 Mod 1
WS 4842	Torpedo Mk 46 Mod 1 Proof Procedure Requirements and Acceptance Criteria

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

National Aerospace standards

NAS 1352-83	Screw, Cap, Socket Head - Undrilled and Drilled, Plain and Self-Locking, Alloy Steel and Corrosion-Resisting Steel, UNRC-3A Thread
NAS 1593	Packing, Preformed - MIL-R-25897 Rubber, 75 Shore, O-Ring

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(Application for copies should be addressed to Aerospace Industries Association of America, Inc., 1775 DeSales Street, Washington, DC 20036.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the grain are classified as follows:

REQUIREMENTS	PARAGRAPH
General Requirements	3.2
Performance Requirements and Product Characteristics	3.3
Environmental Requirements	3.4
Operationsl Service Life	3.5

3.2 General requirements

3.2.1 First article. When specified, a sample shall be subjected to first article inspection (see 6.2.1).

3.2.2 Materials. Unless othersie specified in the contract or purchase order, all materials and treatments of materials used in the manufacture of the grain shall be in accordance with the requirements listed in Dwg 3209534.

3.2.3 Definitions. Definitions of terms are listed in Section 6.3.

3.2.4 Formulation, process, or manufacturing equipment change. No changes of the formulation or manufacturing equipment shall be made without approval of the procuring activity after approval of the first article grains.

3.2.5 Conflicting requirements. In the event of conflict between the documents referenced herein and this specification, the requirements of this specification shall apply.

3.3 Performance requirements and product characteristics. Unless otherwise specified, the grain shall satisfy the following performance requirements and product characteristics when tested as specified in Section 4.

3.3.1 Dimensions and weight

3.3.1.1 Dimensions. Dimensions of the grain shall be in accordance with Dwg 3209534.

3.3.1.2 Weight. The weight of the grain shall not exceed 300 grams.

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3.3.2 Auto-ignition

3.3.2.1 Low temperature. The grain shall be capable of withstanding a temperature of 170°F(77°C) at a relative humidity of less than 30 percent for a period of 2 hours without self-igniting. The grain is required, thereafter, to meet all performance requirements and product characterics of this specification.

3.3.2.2 High temperature. The grain shall be capable of withstanding a temperature of 200°F(93°C) at a relative humidity of less than 30 percent for a period of 8 hours without self-igniting.

3.3.3 Operational requirements

3.3.3.1 Pressure response. The pressure response of the grain shall lie within the curves illustrated in FIGURES 1, 2, and 3 when testing in accordance with 4.5.4.6, low temperature operating parameters; 4.5.4.7, high temperature operating parameters; and 4.5.4.8, operational test parameters.

3.3.3.2 Low pressure operation. The pressure response of the grain shall lie within the curve illustrated in FIGURE 4 when tested in accordance with 4.4.6.18.

3.3.3.3 Combustion chamber temperature. The anticipated flame temperature in the combustion chamber should be approximately 2350°F. The maximum allowable C* (characteristic velocity) equivalent to the 2350°F flame temperature may be calculated in accordance with 3.3.3.3.1. The computed C^*_m based on actual performance data shall be determined in accordance with 4.4.6.17f and this value shall not exceed a maximum value of 3950 ft/sec.

3.3.3.1 Calculation of maximum allowable C*. The maximum allowable C* shall be calculated using the following equation. This value of C* is a theoretical value based on the thermodynamic characteristics of the basis propellant.

$$C^* = \left[\frac{G\ R\ T_c\ o\ c}{\bar{M}}\right]^{1/2} \left[\frac{\delta + 1}{2}\right]^{\frac{\delta + 1}{2(\delta - 1)}} \quad \text{(FT/SEC)}$$

Where:

G_c = 32.174 lbm-ft/lbf-SEC^2 (dimensional constant)
(lbm = pound mass, lbf = pound-force)

R_o = 1543.7 FT-lbf/MOLE - R_o (Gas constant)

T_c = 2810°R (2350°F gas temperature in °R)

δ = Theoretical specific heat ratio

$\bar{M}$ = Theoretical average molecular weight in lbm/mole

NOTE: The manufacturer shall provide the theoretical value of δ and $\bar{M}$ for the propellant as part of the preproduction and batch date package.

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3.4 Environmental requirements. Unless otherwise specified, the grain shall be capable of conforming to the performance requirements and product characteristics of this specification after being exposed to the environmental conditions specified in this section.

3.4.1 Vibration

3.4.1.1 Vibration, ready condition. The grain, mounted as shown in FIGURE 6, shall be capable of meeting performance requirements after being subjected, in the ready condition, to the vibrations defined in TABLE I. The grain shall be vibrated for 180 minutes each in its logitudinal axis and in two mutually perpendicular transverse axes at any temperature within the range of -45°F to +140°F (-42.8°C to +60°C).

3.4.1.2 Vibration, packaged condition. The grain shall be capable of meeting performance requirements after being subjected, in the packaged condition, to the vibrations shown in TABLE I.

TABLE I. Vibration schedule

Condition	Double amplitude (in. ± 10%)	Acceleration (g's ± 10%)	Frequency (Hz)
Packaged	---	1.0	5 (minimum)
Ready	0.2	---	5 - 17
	---	3.0	17 - 50
	0.036	---	40 - 74
	---	10	74 - 500

3.4.2 Acceleration, ready condition. The grain, mounted as shown in FIGURE 6, shall be capable of meeting performance requirements after being subjected, in the ready condition, to the accelerations defined in TABLE II for a duration of one minute for each magnitude.

TABLE II. Acceleration schedule

Magnitude (g's ± 10%	Acceleration direction (See FIGURE 6)
25	Longitudinal axis, fwd
35	Longitudinal axis, aft

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3.4.3 Shock, ready condition. The grain, mounted as shown in FIGURE 6, shall be capable of meeting performance requirements after being subjected, in the ready condition, to the shock loads defined in TABLE III.

TABLE III. Shock schedule

Peak Magnitude (g's ± 10%)	Duration (ms)	Impact axis	Direction of deceleration force	Number of impacts
300	0.5 to 1.5	Longitudinal	Aft	3
85	0.5 to 1.5	Transverse A	Optional	3
85	0.5 to 1.5	Transverse B	Optional	3
45	18.0 to 20.0	Longitudinal	Fwd	3
60	8.0 to 10.0	Longitudinal	Aft	3
60	8.0 to 10.0	Longitudinal	Fwd	3
60	8.0 to 10.0	Transverse A	Optional	3
60	8.0 to 10.0	Transverse B	Optional	3

3.4.4 Temperature and humidity

3.4.4.1 Nonoperating

a. Temperature. The grain shall be capble of meeting performance requirements subsequent to withstanding, in the nonoperating condition, environmental temperatures between -45°F and +140°F (-42.8°C and +60°C).

b. Humidity. The grain shall be capable of meeting performance requirements subsequent to withstanding, in the nonoperating condition, an environment of at least 95 percent relative humidity at (90°F) 32.2°C for a minimum period of 12 hours.

3.4.4.2 Ready, operating. The grain after being conditioned for 3 hours at any temperature between 0°F and +140°F (17.8°C and +60°C) shall be capable of meeting performance requirements at any temperature between 0°F and +140°F (-17.9°C and +60°C).

3.4.5 Jolt. The grain, mounted as shown in FIGURE 6, shall be capable of withstanding the jolt test requirements of Test No. 101 of MIL-STD-331. It is not required that the grain be operable after testing. The criteria by which the samples are judged to have withstood the testing are that (1) no explosive elements shall explode and (2) no parts shall be broken, be deformed, be displaced, come apart, or arm, in such a manner as to make the assembly unsafe to handle or dangerous to use, as determined by examination.

3.4.6 Jumble. The grain, in the nonoperating condition, shall be capable of withstanding the jumble test requirements of Test No. 102 of MIL-STD-331. It is not required that the grain be operable after testing. The criteria by which the samples are judged to have withstood the testing are that (1) no explosive elements shall explode and (2) no parts shall be broken, be deformed, be displaced, come apart, or arm in such a manner as to make the assembly unsafe to handle or dangerous to use, as determined by examination.

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3.4.7 Drop. The grain, in the nonoperating condition, shall be capable of withstanding the drop test requirements of Test No. 103 of MIL-STD-331. It is not required that the grain be operable after testing. The criteria by which the samples are judged to have withstood the testing are that (1) no explosive elements shall explode and (2) no parts shall be broken, by deformed, be displaced, come apart, or arm, in such a manner as to make the assembly unsafe to handle or dangerous to use, as determined by examination.

3.4.8 Accelerated storage life

3.4.8.1 Packed and packaged condition. The grain, when packed with cushions and spacer that have been exposed to a minimum of 95 percent relative humidity at 110°F (43°C) for 24 hours, shall be capable of withstanding an environmental temperature of 140°F (60°C) for a period of 30 days. Thereafter the grain shall be capable of meeting the requirements of 3.4.8.2.

3.4.8.2 Ready condition. Subsequent to exposure to a minimum of 95 percent relative humidity at 90°F (32.2°C) for a period of 12 hours, the grain in the sealed ready condition shall be capable of withstanding an environmental temperature of 140°F (60°C) for a period of 45 days. Thereafter the grain shall be capable of operating in accordance with 3.3.3.

3.4.9 Service capability. The grain, when installed in Torpedo Mk 46 Mod 1, shall be capable of initiating the normal starting sequence subsequent to the launching of the torpedo from a high speed, fixed wing aircraft.

3.5 Operational service life. The grain shall have a service life of one duty cycle.

3.6 Workmanship. The workmanship and finish shall be of sufficiently high grade to ensure satisfactory operation consistent with the requirements of this specification. The assembly shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship exhibited in any approved first article sample, shall be determinative of the requirements of workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the grain shall be classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

c. Periodic production inspection (see 4.6)

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4.3 Sampling

4.3.1 First article inspection samples. Two groups, A and B, of first article sample grains shall be subjected to first article inspection requirements at an activity designated by the procuring activity.

4.3.1.1 Group A first article inspection sample. A first article sample of five grains manufactured in accordance with 3.2 shall be subjected to first article testing detailed in 4.4.2. Acceptance of the first article sample shall be based on no defects in the sample (see 4.4.2). Further production of the grains by the contractor, prior to approval of the first article sample, shall be at the contractor's risk.

4.3.1.2 Group B first article sample. A first article sample of 25 grains manufactured in accordance with 3.2 shall be subjected to the first article tests, listed in TABLE IV and detailed in 4.4.6.1 through 4.4.6.18. Acceptance of the first article sample shall be based on no defects in the sample (see 4.4.8). Further production of the grains by the contractor, prior to approval of the first article sample, shall be at the contractor's risk.

TABLE IV. Sample Group B, first article test sequence

Tests	Paragraph reference	Test sequence												
		A	B	C	D	E	F	G	H	I	J	K	L	M	N
Inspection	4.4.6.1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Vibration, ready condition	4.4.6.2	2													
Acceleration, ready condition	4.4.6.3		2												
Shock, ready condition	4.4.6.4		3												
Temperature and humidity, nonoperating condition	4.4.6.5			2	2										
Low temperature operating	4.4.6.6			3					4						
High temperature operating	4.4.6.7				3			4							
Auto-ignition, low temperature	4.4.6.8					2	2								
Auto-ignition, high temperature	4.4.6.9									2					
Vibration, packed and packaged	4.4.6.10					3									
Shock, packed and packaged	4.4.6.11					4									
Jolt	4.4.6.12												2		
Jumble	4.4.6.13													2	
Drop	4.4.6.14														2
Accelerated storage life, packed and packaged condition	4.4.6.15							2	2			2			
Accelerated storage life, ready condition	4.4.6.16							3	3			3			
Normal Pressure operational test	4.4.6.17	3	4			5	3								
Low pressure operational test	4.4.6.18										2	4			

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4.3.2 Quality conformance inspection sample. The sampling plans and procedures set forth in MIL-STD-105 shall be used in the determination of the acceptability of the lot of grains inspected as specified in 4.5.4.1. In addition, a quantity of 50 sample grains from each lot shall be subjected to the tests specified in 4.5.4.2 through 4.5.4.9 in accordance with TABLE V.

TABLE V. Quality conformance inspection test sequence

Tests	Paragraph reference	Sample and test sequence							
		1,2	3,4	5,6	7,8	9-18	19-28	29-38	39-50
Vibration, ready condition	4.5.4.2	X							
Jolt	4.5.4.3		X						
Jumble	4.5.4.4			X					
Drop	4.5.4.5				X				
Low temperature operating	4.5.4.6						X		
High temperature operating						X			
Normal pressure operational test	4.5.4.8	X							X
Low pressure operational test	4.5.4.9							X	

4.3.3 Periodic production sample. Periodic production samples shall be selected from units that have satisfactorily passed the quality conformance inspections of 4.5.4.1. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

a. Upon initiation of production, randomly select six units from each batch of grains. Upon satisfactory compliance of samples selected from three consecutive batches, proceed to 4.3.3.b.

b. Randomly select six units from every fifth batch of grains. Continue testing in accordance with 4.6 for the balance of the contract or until a defective unit is found. When a defective unit is found in any batch, acceptance of all units in that batch shall be withheld (see 6.2.1).

4.4 First article inspection

4.4.1 Tests and examinations. The Group A first article sample of five grains shall be subjected to the tests delineated in 4.4.2. The Group B first article sample of 25 grains shall be subjected to the tests delineated in 4.4.3.

4.4.2 Group A first article tests. The five grains shall be installed in a Mk 46 Mod 1 exercise configuration torpedo in accordance with WD 30369, and the launching plan, Appendix D, of WS 4842. Acceptance of the grain shall be based on the condition that, after being ignited, the grain shall initiate the propulsion system operation, and ignite the Otto Fuel, as evidenced by the torpedo exercise section data recording system.

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4.4.3 Group B first article tests. The first article sample of 25 grains shall be inspected in accordance with TABLE IV. Test sequences A through K shall be conducted on two grains each, and sequences L through N shall be conducted on one grain each.

4.4.4 Test conditions. Unless otherwise specified, the grain shall be subjected to the tests in 4.4.3 under the following conditions:

a.	Temperature:	Room ambient, 32°F to 98°F (0°C to 36.7°C)
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.4.5 Test equipment and setup. In addition to the items listed in 4.5.3, the following items are required for the first article tests:

4.4.5.1 Vibration equipment with temperature chamber. The vibration equipment with temperature chamber shall have the following minimum characteristics:

a. Frequency range, logarithmic sweep: 5 to 500 Hertz (Hz)

b. Test specimen: 5 pounds

c. Acceleration and displacement (ready condition):

5 to 17 Hz	0.2 inch double amplitude
17 to 40 Hz	3.0 gravitational units (g's)
40 to 74 Hz	0.036 inch double amplitude
74 to 500 Hz	10 g's

d. Acceleration (packed and packaged):

5 to 60 Hz	1 g

e. Automatic cycling time range: 0 to 15 minutes (optional).

f. Temperature chamber:

(1)	Range:	-45°F to +140°F (-42.8°C to +60°C)
(2)	Accuracy:	±5°F (±2.8°C)

g. Interior size: 4 inches x 4 inches x 6 inches minimum.

4.4.5.2 Acceleration equipment. The acceleration equipment shall have the following minimum characteristics: 35 g's with a 5-pound test specimen.

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4.4.5.3 Shock equipment. The shock equipment shall be capable of accepting a 5-pound test specimen and shall provide ranges of :

a. 300 g's for 0.5 milliseconds (ms) to 1.5 ms
b. 60 g's for 8.0 ms to 10.0 ms
c. 45 g's for 18.0 ms to 20.0 ms

4.4.5.4 Temperature and humidity conditioning equipment. One or more equipments may be utilized to provide the following characteristics:

a. Temperature: -45°F to +250°F (-42.8°C to +121°C); with an accuracy of ±5°F (±2.8°C)

b. Relative humidity: 95 percent minimum at 110°F (43°C) with an accuracy of ±5°F (±2.8°C)

c. Interior size: 4 inches x 4 inches x 6 inches minimum

NOTE: The equipment cited in 4.4.5.4 may be used for both nonoperating and ready/operating temperature and humidity tests.

4.4.6 Test procedure

NOTE: Tests on the grain, in the nonoperating condition, shall be performed without packaging or packing.

4.4.6.1 Inspection. The grain shall be subjected to the inspection detailed in 4.5.4.1.

4.4.6.2 Vibration, ready condition. The grain, in the ready condition, shall be mounted in a manner equivalent to that shown in FIGURE 6. The resonant modes in each of the 3 axes shall be determined by varying the frequency of applied vibration slowly from 5 Hz to 500 Hz. The grain shall then be subjected to the resonance and cycling vibration in each axis, in conformance with the time shown in TABLE VI. If more than four resonances are encountered on any one axes, the four most severe resonances shall be chosen for the test. The resonant and cycling period of 180 minutes for each axis shall be divided into 3 equal parts, which are conducted at -45° ±5°F (-42.8° ±2.8°C), room temperature, and +140° ±5°F (+60° ±2.8°C), respectively. For cycling, a logarithmic frequency sweep rate shall be used if available; otherwise, other cycling rates of frequency change may be used. The sweep rate shall be such that 15 minutes actual vibration time is required to proceed from the lower to higher and back to the lower extremity of the frequency range in accordance with the accelerations or double amplitudes defined in TABLE I.

TABLE VI. Resonance and cycling schedule

No. of resonances	Time at resonance at each temperature in each axis, and at each resonance (min)	Cycling time at each temperature in each axis (min)
0	0	60
1	10	50
2	10	40
3	10	30
4	10	20

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4.4.6.3 Acceleration, ready condition. The grain, in the ready condition, mounted as shown in FIGURE 6, shall be subjected to the accelerations defined in TABLE II for a duration of 1 minute for each magnitude. Acceleration shall be measured by an accelerometer mounted on the combustion chamber at a point approximately coinciding with the longitudinal center of the grain.

4.4.6.4 Shock, ready condition. The grain, in the ready condition, mounted as shown in FIGURE 6, shall be attached to a shock test machine. Shock magnitudes and durations shall be measured by an accelerometer mounted on the mounting fixture. Tests on the grain shall be performed in accordance with TABLE III. Test sequence is optional.

4.4.6.5 Temperature and humidity, nonoperating condition. The grain shall be subjected, in the nonoperating condition to temperature and humidity cycling in the following manner. The grain shall be placed in a temperature chamber which has been stabilized at an environmental temperature of -45° ±5°F (-42.8° ±2.8°C). The grain shall remain in that environment for 6 hours. (As an alternative procedure, the grain may be placed in the chamber, with the chamber at room ambient temperature. Chamber temperature shall then be reduced to -45° ±5°F (-42.9° ±2.8°C) within 2 hours maximum. The grain shall thereafter remain in the chamber for 4 hours.) The grain shall then be subjected to an environmental temperature of +140° ±5°F (+60° ±2.8°C). Maximum transition time from the low temperature to the high temperature shall be 2 hours. The grain shall remain in the high temperature for a minimum of 4 hours. This cycle shall be performed 3 times; maximum transition shall be 2 hours. The grain shall subsequently be subjected to an environment of 90° ±5°F (±2.8°C) at a minimum relative humidity of 95 percent for a minimum of 12 hours (±15 minutes). The grain may be wiped dry. Drying by an air blast or oven is not permitted.

4.4.6.6 Low temperature operating. Before testing, the grain, installed in its test chamber, FIGURE 5, in the ready condition, shall be conditioned at a temperature of 0° ±5°F (-17.8° ±2.8°C) for a period of 3 hours. Within 15 minutes after removal from the low temperature environment, the grain shall be operated in accordance with and meet the requirements of 4.4.6.17, except that the pressure-time curve shall conform to FIGURE 2.

4.4.6.7 High temperature operating. Before testing, the grain, installed in its test chamber, FIGURE 5, in the ready conditon, shall be conditioned at a temperature of +140° ±5°F (+60° ±2.8°C) for a period of 3 hours. Within 15 minutes after removal from the high temperature environment, the grain shall be operated in accordance with and meet the requirements of 4.4.6.17, except that the pressure-time curve shall conform to FIGURE 3.

4.4.6.8 Auto-ignition, low temperature. The grain, in the nonoperating condition, shall be subjected for 2 hours to a temperature of +170° ±10°F (+77° (±5.6°C) at a relative humidity of less than 30 percent. Combustion shall not occur. The grain shall, subsequently, be capable of meeting subsequent scheduled tests and operational test parameters.

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4.4.6.9 Auto-ignition, high temperature. The grain, in the non-operating condition, shall be subjected for 8 hours to a temperature of +200° ±20°F (+93° ±11.1°C) at a relative humidity of less than 30 percent. Combustion shall not occur. The grain need not be capable of operation after the auto-ignition high temperature test.

4.4.6.10 Vibration, packed and packaged. Place the packed and packaged grain on the vibration platform, fenced on all sides so that the pack cannot fall off the platform. Starting at 5 Hz, gradually increase the frequency until the pack just begins to leave the platform. At this point, run the test for one hour.

4.4.6.11 Shock, packed and packaged. The grain, packed and packaged in accordance with Section 5 shall be subjected to free-fall drop tests. The grain in its container shall be dropped from a height of 36 inches. A total of four edgewise drops shall be made. Two drops, each, shall be amde on the bottom and top edges at points 180 degrees apart. In addition, flat drops shall be made, one each, on the bottom and one the top. The drops shall be made on an impact surface of essentially non-yielding material such as stone or concrete.

4.4.6.12 Jolt. The grain, mounted as shown in FIGURE 6, shall be jolt-tested in accordance with and shall meet the requirements of Test No. 101 of MIL-STD-331. The grain need not be capable of operation after the jolt test.

4.4.6.13 Jumble. The grain, in the nonoperating conditon, shall be jumble-tested in accordance with and shall meet the requirements of Test No. 102 of MIL-STD-331. The grain need not be capable of operation after the jumble test.

4.4.6.14 Drop The grain. in the nonoperating condition, shall be drop-tested in accordance with and shall meet the requirements of the method prescribed by Test No. 103 of MIL-STD-331. The grain shall impact on the end opposite the carbon spacer. The grain need not be capable of operation after the drop test.

4.4.6.15 Accelerated storage life, packed and packaged condition. The cushion and spacer shall be removed from the packaging container and subjected to a minimum of 95 percent relative humidity at 110° ±5°F (43° ±2.8°C) for a period of 24 hours (±15 min). Immediately thereafter, the grain shall be packed using the conditioned cushions and spacer and the container shall be sealed. The container shall be subjected to an environmental temperature of 140° ±5°F (60° ±2.8°C) at ambient atmospheric pressure condition for a period of 30 days.

4.4.6.16 Accelerated storage life, ready condition. The grain in the nonoperating condition shall be subjected to a minimum of 95 percent relative humidity at 90° ±5°F (32.2° ±2.8°C) for a period of 12 hours (±15 min). Immediately thereafter the grain shall be placed in the ready condition except that the combustion chamber or housing must be sealed with a plug having the mating requirements of the test fixture shown in FIGURE 7. The grain may be wiped dry prior to installation. The combustion chamber or housing shall then be subjected to an environmental temperature of 140° ±5°F (60° ±2.8°C) at ambient atmospheric pressure for a period of 45 days.

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4.4.6.17 Normal pressure operational test

a. The grain and test chamber shall be enclosed by a barrier to provide protection for operating personnel.

b. Mount the grain in the test fixture shown in FIGURE 5. The orifice should be as specified in FIGURE 7, Orifice A. Instrumentation shall be in accordance with FIGURE 8 or 9,

c. Unless otherwise specified, condition the grain installed in the test fixture in the ready condition at 60° ±5°F (15.6° ±2.8°C) for a period of 3 hours.

d. Within the 15 minutes after removing the grain (in its test fixture) from the temperature conditioning chamber, initiate grain firing by closing the switch illustrated in FIGURE 8 or 9. The igniter should fire within 21 milliseconds (ms).

e. After the test, plot the combustion chamber pressure versus time obtained from either digital data acquisition or analog oscillograph recordings. The pressure-time curve shall fall within the ranges shown in FIGURE 1.

f. The computed C_m^* shall be determined by using integral of the area, bounded by the pressure-time curve specified in 4.4.6.17(e), and applied in the following equation:

$$C_m^* = \frac{A_t G_c K \int_{t_1}^{t_2} Pdt}{W_P} \quad \text{(FT/SEC)}$$

Where:

A_t = Average area of throat based on throat size before and after the test.

G_c = 32.174 lbm-ft/lbf-sec (dimensional constant)

K = .95 (coefficient of test accuracy)

$\int_{t_1}^{t_2} Pdt$ = Area in psig-sec under the pressure-time curve measured in the time interval (t_2 - t_1)

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$t_2 - t_1$ = Time interval in seconds from time (t_1) to attainment of 10% of maximum pressure in the rise portion of the pressure-time-curve to the time (t_2) when the same pressure (10% level) has been attained in the decay portion of the pressure-time curve (ref. MIL-STD-292).

W_p = Weight in lbm of the propellant grain (sustainer propellant portion only) of the tested grain. The manufacturer shall provide the value of W_p for each grain to be tested by serial number, as part of the pre-test data package.

C_m^* = The computed characteristic velocity (C^*) in ft/sec shall not exceed the maximum allowable C^* determined in accordance with 3.3.3.3.1 or 3950 ft/sec, whichever is the lesser value.

4.4.6.18 Low pressure operational test

a. The grain and test chamber shall be enclosed by a barrier to provide protection for operating personnel.

b. Mount the grain in the test fixture as shown in FIGURE 9. The orifice shall be as specified in FIGURE 7, Orifice "B". The electrical power required for firing the igniter shall be in accordance with FIGURE 8 or 9.

c. The grain, installed in the combustion chamber in the ready condition, shall be conditioned at a temperature of 60° ±5°F (15.6° ±2.8°C) for a period of 3 hours.

d. Within 15 minutes after removing the grain from the temperature conditioning chamber, initiate grain firing by closing the switch illustrated in FIGURE 8 or 9.

e. After the test, plot the combustion chamber pressure versus time obtained from either digital data acquisition or oscillograph record. The pressure-time curve shall fall within the ranges shown in FIGURE 4.

4.4.7 Test data sheets. All test data shall be recorded on a data sheet. Data sheets and oscillographic records or digital data acquisition printouts and plots shall be maintained as part of the inspection records. Each trace on the analog or digital records shall be identified as to function and calibration. The applicable data for each sample grain shall be recorded on the data sheets. Data sheets shall contain but are not limited to the following information:

APPLICABLE INSPECTION DATA, GRAIN

a. Date.

b. Grain serial number.

c. Instrumentation and calibration date.

d. Dimensions.

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e. Computed C_m^* (characteristic velocity) (ft/sec). Maximum allowable C* (Calculated or 3950 ft/sec, whichever is smaller).

f. Combustion chamber pressure in psig (from either digital data acquisition or oscillographic record), plot of pressure response curve.

g. Operating temperature.

h. Auto-ingnition temperature and length of time subjected to temperature.

i. Accelerated storage life temperature and number of days subjected to environmental temperature.

j. Propellant weight (sustainer propellant portion only).

4.4.8 Consequence of defect or failure. Failure of a grain to successfully pass any of the test in 4.4.6.1 through 4.4.6.18 shall be cause for rejection of the first article (see 6.2).

4.5 Quality conformance inspection tests

4.5.1 Tests and examinations. Grains produced under the contract or purchase order shall be subjected to the following tests and examinations.

a. Each grain in the lot shall be subjected to the test detailed in 4.5.1a.

b. A sample of grains shall be selected using the plans and procedures set forth in MIL-STD-105, Inspection Level II, and subjected to the examinations in 4.5.4.1(b), (c), and (d). An AQL of 1.0 shall be used for 4.5.4.1(b) and (c), and an AQL of 2.5 shall be used for 4.5.4.1(d).

c. A sample of 50 grains shall be selected from each lot of grains submitted. The sample shall be subjected to the tests and examinations in 4.5.4.2 through 4.5.4.9 in accordance with TABLE V.

4.5.2 Test condition. Unless otherwise specified, the grain shall be subjected to the quality conformance tests under the following conditions:

a.	Temperature:	Room ambient, 32°F to 98°F (0°C to 36.7°C
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.5.3 Test equipment and setup. The following items of equipment are required.

a.	Test fixture:	FIGURE 5

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b. Igniter: See MIL-I-82437

c. Power supply: See MIL-I-82437

d. Pressure transducer:

(1)	Range:	0 to 3000 psig
(2)	Linearity:	±1 percent full scale maximum
(3)	Resolution:	±0.5 percent full scale maximum
(4)	Repeatability:	±0.5 percent full scale maximum
(5)	Hysteresis:	±1 percent full scale minimum
(6)	Proof pressures:	150 percent full scale minimum
(7)	Tempeature range:	0°F to 180°F (-17.8°C to 82°C minimum

e. Oscillographic recorder (two channels minimum; permanent recording):

(1)	Frequency response:	0 to 20 Hz ±5 percent minimum
(2)	Chart speed:	2.0 inches per second minimum
(3)	Recording paper or film:	6-inch width minimum
(4)	Time base:	0.10 second markers

f. Vibration equipment with temperature chamber. The vibration equipment shall be in accordance with 4.4.5.1.

g. Digital data acquisition system:

(1)	Frequency response:	0-2000 Hz at -3 db
(2)	Sampling rate:	19,200 sps max
(3)	Resolution:	12 bits + sign (±4095)
(4)	Non-linearity:	Less than ±0.08%, 3 Sigma best straight line

4.5.4 Test procedures

4.5.4.1 Configuration examination. (refer to Dwg 3209534)

a. Visual. Each grain shall conform to the requirements of Note 4 of the drawing.

b. Materials. The spacer and pad materials shall conform to the requirements of the drawing.

c. Dimensions. The grain diameter and length shall conform to the requirements of the drawing.

d. Marking. The marking shall be in accordance with the drawing.

4.5.4.2 Vibration, ready condition. The grain shall be subjected to and meet the vibration, ready condition test requirements specified in 4.4.6.2.

4.5.4.3 Jolt. The grain shall be subjected to and meet the requirements specified in 4.4.6.12.

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4.5.4.4 Jumble. The grain shall be subjected to and meet the test requirements specified in 4.4.6.13.

4.5.4.5 Drop. The grain shall be subjected to and meet the test requirements specified in 4.4.6.14.

4.5.4.6 Low temperature operating. The grain shall be subjected to the procedures of 4.4.6.6 and shall meet its requirements.

4.5.4.7 High temperature operating. The grain shall be subjected to the procedures of 4.4.6.7 and shall meet its requirements.

4.5.4.8 Normal pressure operational test. The grain shall be subjected to the procedures of 4.4.6.17 and shall meet its requirements. (Exhaust gas temperatures need not be recorded).

4.5.4.9 Low pressure operational test. The grain shall be subjected to the test of 4.4.6.18 and shall meet its requirements. Computed characteristic velocity (C_m^*) for each grain tested shall be determined by the method defined in 4.4.6.17(f) and shall not exceed the maximum allowable C_m^* or 3950 ft/sec, whichever value is smaller.

4.5.5 Test data sheets. All test data shall be recorded on a data sheet. Data sheets and oscillographic records or digital data acquisition printouts and plots shall be maintained as part of the inspection records. Each trace on the analog or digital records shall be identified as to function and calibration. The applicable data for each sample grain shall be recorded on the data sheets. Data sheets shall contain, but are not limited to, the following information:

QUALITY CONFORMANCE INSPECTION DATA, GRAIN

a. Date.

b. Grain serial number.

c. Dimensions.

d. Combustion chamber pressure in psig (from either digital data acquisition or oscillographic record), plot of pressure response curve.

e. Operating temperature.

f. Computed C_m^* (characteristic volocity) (ft/sec). Maximum allowable C* (calculated or 3950 ft/sec, whichever is smaller).

g. Propellant weight (sustainer propellant portion only).

4.5.6 Consequence of defect or failure. Failure of a grain in the acceptance sample of 50 grains to successfully pass any of the tests in 4.5.4.2 through 4.5.4.9 shall result in rejection of the lot. Failure of a grain to successfully pass the test of 4.5.4.1(a) shall result in rejection of the grain.

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4.6 Periodic production inspection

4.6.1 Tests and examinations. Periodic production samples shall be subjected to examination and tests in accordance with TABLE VII. Test sequences A through C shall be conducted on two grains each.

TABLE VII Periodic test sequence

Tests	Paragraph reference	Test sequence A	B	C
Inspection	4.6.4.1	X	X	X
Shock, ready condition	4.6.4.2	X		
Auto-ignition, high temperature	4.6.4.3			X
Temperature and humidity, nonoperating condition	4.6.4.4		X	
Normal pressure operational test	4.6.4.5	X	X	

4.6.2 Test conditions. Unless otherwise specified, the grain shall be subjected to the periodic tests under the following conditions:

a. Temperature: Room ambient, 32°F to 98°F (0°C to 36.7°C)

b. Altitude: Normal ground atmospheric pressure

c. Vibration: None

d. Humidity: Room ambient

4.6.3 Test equipment and setup. The test equipment and setup for the periodic tests shall be in accordance with 4.4.5.

4.6.4 Test procedures

NOTE: Tests on the grain, in the nonoperating condition, shall be performed without packaging or packing.

4.6.4.1 Inspection. Inspection of the grain shall be performed in accordance with 4.5.4.1(a) through (d).

4.6.4.2 Shock, ready condition. Shock testing of the grain, in the ready condition, shall be perfomred in accordance with 4.4.6.4.

4.6.4.3 Auto-ignition, high temperature. The grain shall be subjected to the auto-ignition, high temperature test in accordance with 4.4.6.9.

4.6.4.4 Temperature and humidity, nonoperating condition. Temperature and humidity testing of the grain, in the nonoperating condition, shall be performed in accordance with 4.4.6.5.

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4.6.4.5 Normal pressure operational test. The grain shall be subjected to the normal pressure operational test in accordance with 4.4.6.17.

4.6.5 Test data. Test data shall be recorded in accordance with 4.4.7, as appropriate, and maintained as part of the inspection record.

4.6.6 Consequence of defect or failure. Failure of the grain to successfully pass the tests in 4.6.4.1 through 4.6.4.5 shall be cause for withholding acceptance of the units on hand (see 6.2.1).

4.6.7 Packaging, packing, palletizing, and marking inspection. The grains shall be packaged, packed, palletized and marked in accordance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging of the grain shall be in accordance with the instructions contained in DL 2541709.

5.2 Packing. The grains, packaged in accordance with 5.1, shall be packed in accordance with the instructions contained in DL 2541708.

5.3 Palletizing. The grains, packed in accordance with 5.2, shall be palletized in accordance with MIL-STD-147, Load Type XIII. A maximum of 80 packs (640 grains) shall constitute a pallet load.

5.4 Marking. The packages and packs shall be marked in accordance with the applicable packaging drawings (see 5.1 and 5.2). Palletized loads shall be marked in accordance with MIL-STD-129 and shall include the following information:

"Propellant Explosive Solid Class B
CG II-A"

6. NOTES

6.1 Intended use. The grain is intended for use in a torpedo propulsion system to ignite a liquid propellant.

6.2 Ordering data

6.2.1 Procurement requirements. Preocurement documents should specify the following:

a. Title, number, and date of thie specification.

b. Quantity of grain.

c. Requirement for a first article sample (3.2.1).

d. Any exceptions to this sepcification (3.2.2, 3.4, 4.1, 4.4.4, 4.4.6.17c, 4.5.2).

e. Assigned activity for preproduction and periodic production tests. (4.3.1, 4.3.3).

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f. Action to be taken in the event of defect or failure. (4.3.3.b, 4.4.8, 4.6.6).

g. The the safety precaution requirements of the Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material, DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by the government activity, the safety precaution requirements of Ammunition and Explosives Ashore, OP 5, should be made applicable.

6.3 Definitions

6.3.1 Ready. The condition of the grain when fully assembled in its housing or equivalent, prior to application of igniter firing current.

6.3.2 Operating. The condition of the grain when fully assembled in its housing, or equivalent, from the instant of application of firing current to the termination of generation of grain combustion products.

6.3.3 Nonoperating. The condition of the grain when stored, transported, or handled.

6.3.4 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplidude may refer to displacement or to acceleration . The displacement is given in inches peak-to-peak or the corresponding maximum acceleration in gravitational units. The frequency is given in Hertz (Hz).

6.3.5 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in gravitational units.

6.3.6 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a buildup to 90 percent of peak force in less than 0.010 second. Shock is described as a pulse in units of gravitational acceleration, duration in milliseconds, and shock wave-form equivalent to one-half sine wave. Duration is defined as the interval between the points of zero amplitude, from which the shock wave form builds up and to which it decays.

6.3.7 Lot. A group of grains made from the same material with the same equipment at approximately the same time and submitted for inspection at one time. All grains in any one lot must be manufactured from a single batch.

6.3.8 Batch. A group of grains made from the same material with the same equipment at approximately the same time and submitted for insepection at one time. A batch may be subdivided into a number of lots, if desired, for inspection purposes.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Navy - OS

Preparing Activity:
Navy - OS
Project No. 1355-N398

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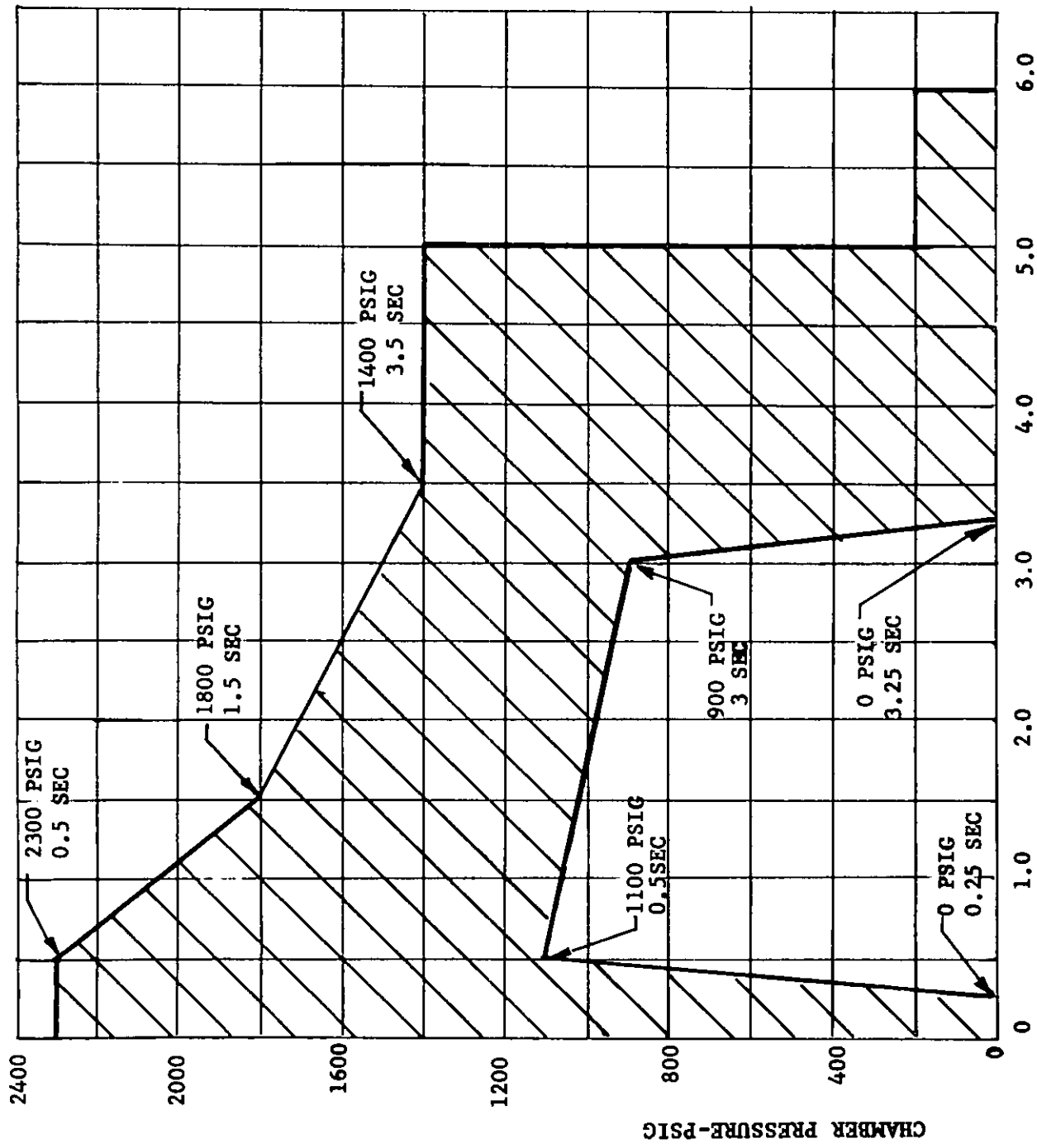


FIGURE 1. Pressure-time curve, 60°F.

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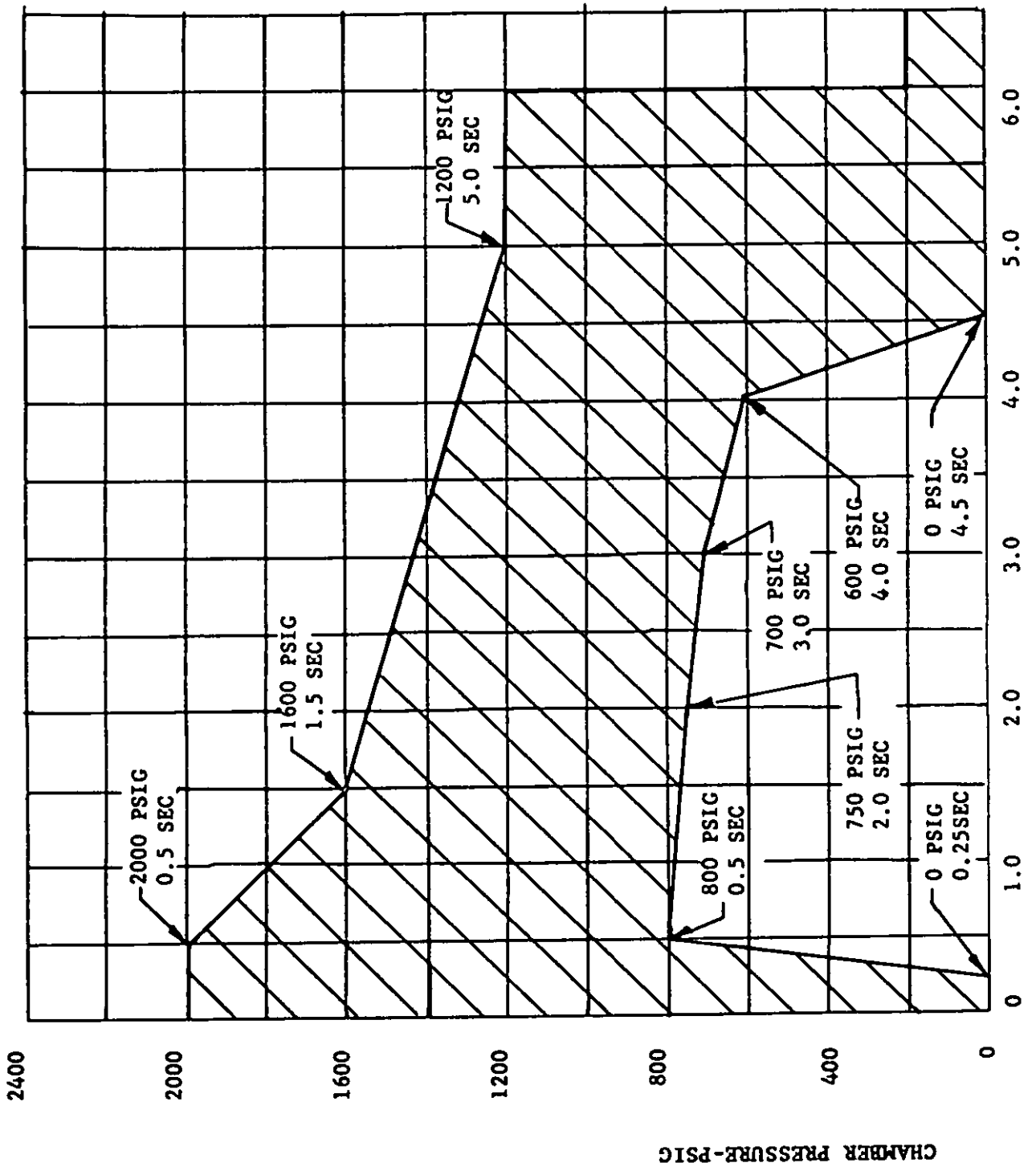


FIGURE 2. Pressure-time curve, 0°F.

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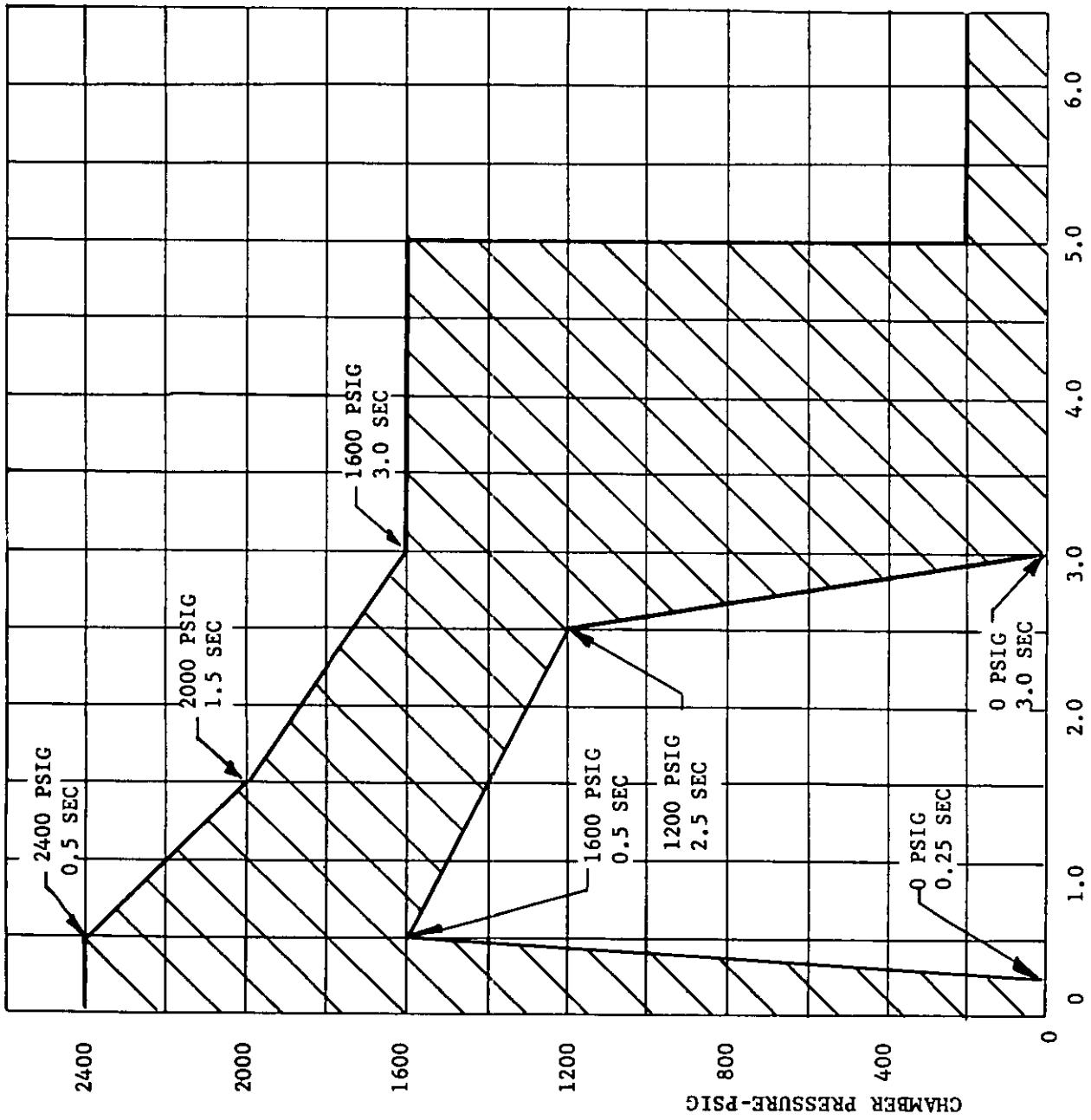


FIGURE 3. Pressure-time curve, 140°F.

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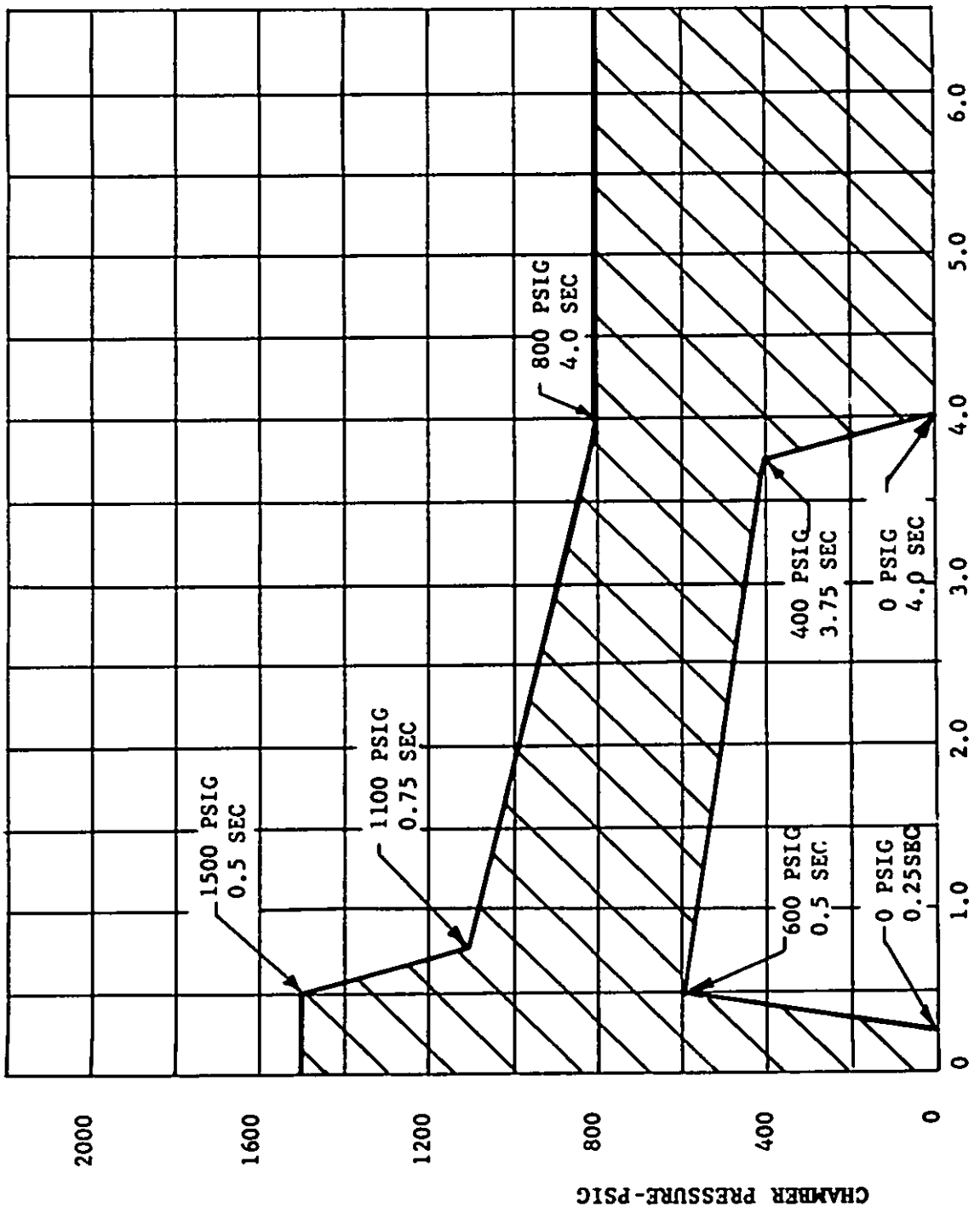


FIGURE 4. Pressure-time curve, 60°F, large orifice.

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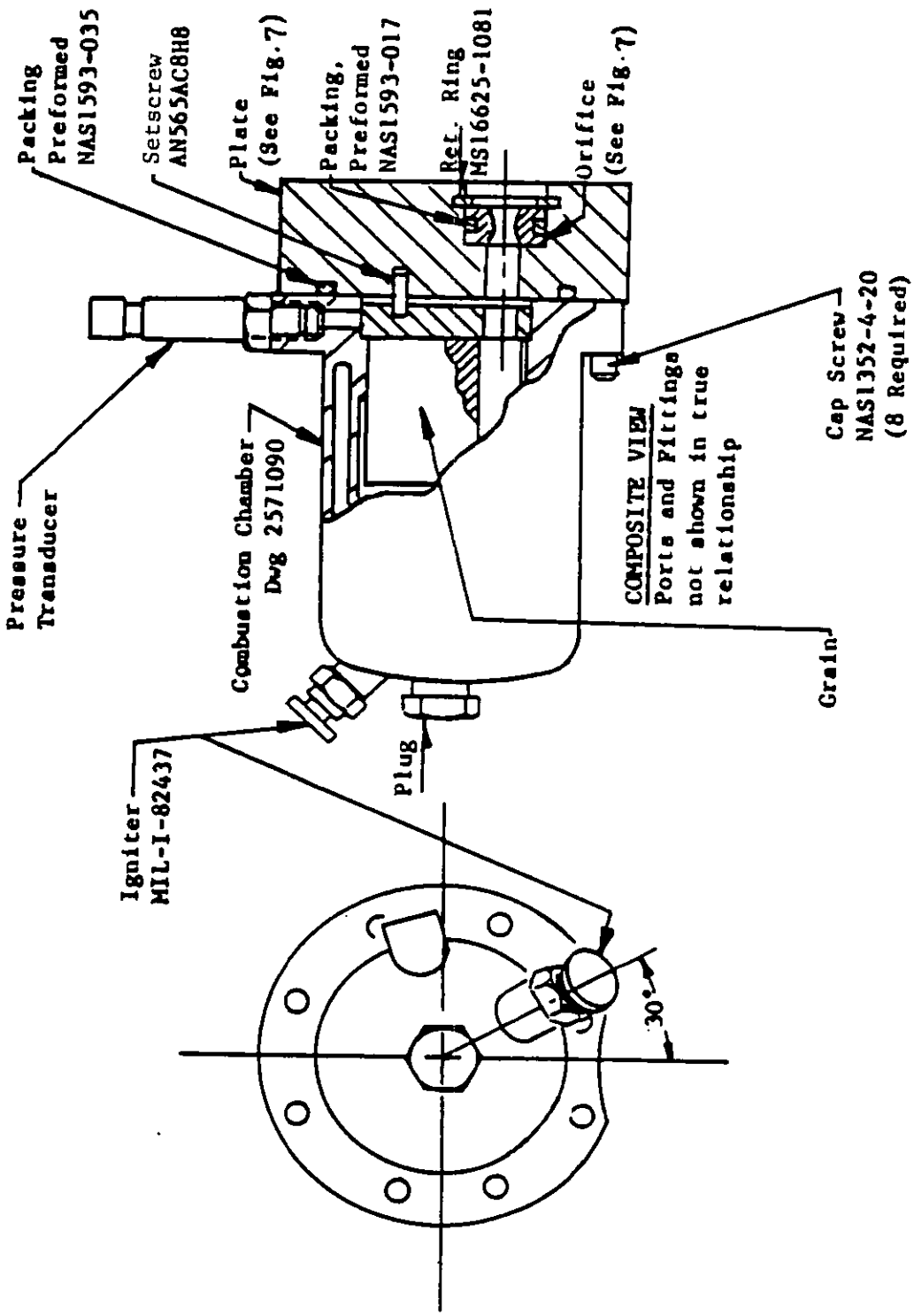
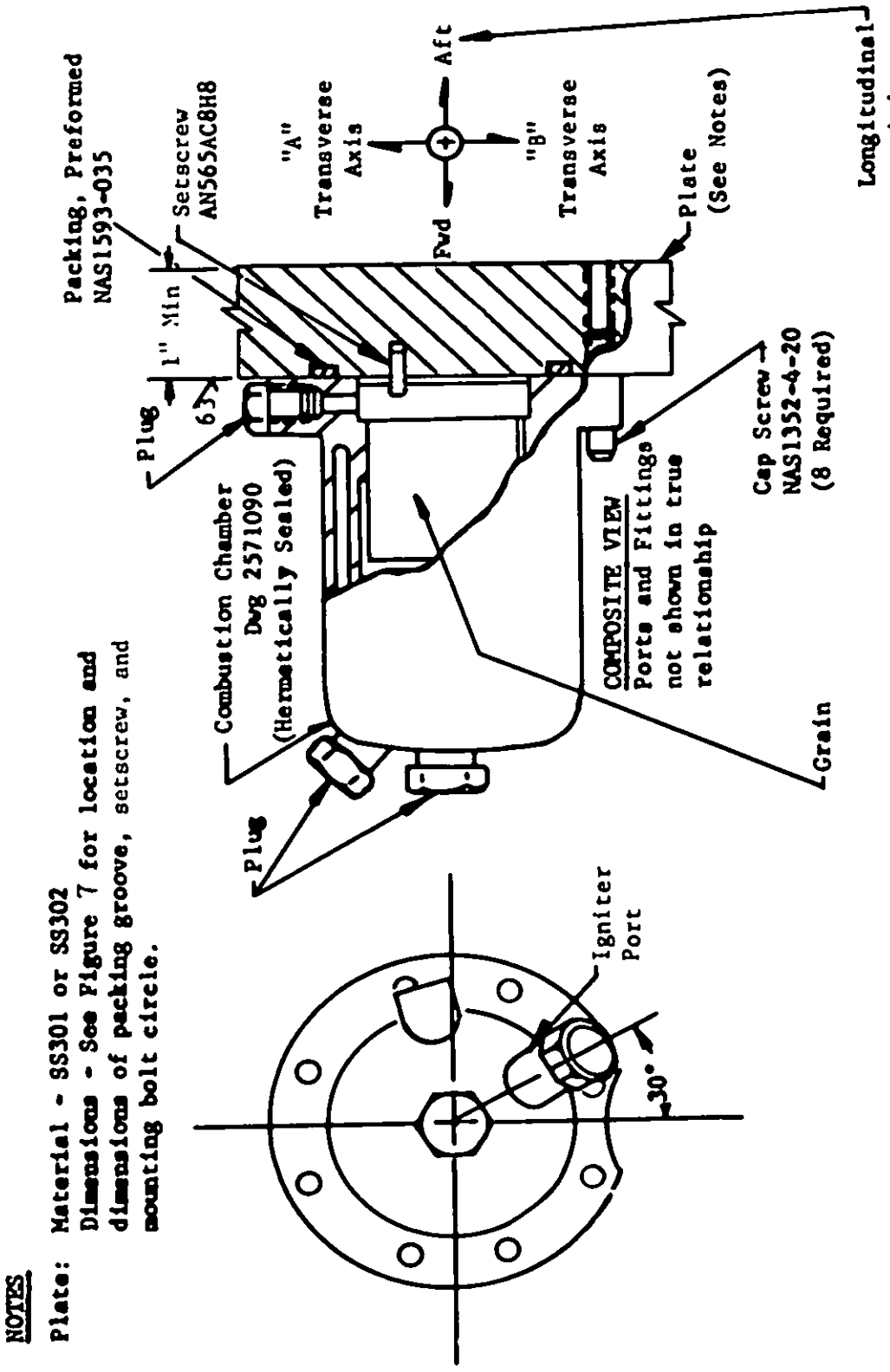


FIGURE 5. Operational test assembly

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NOTES

Plate: Material - SS301 or SS302
Dimensions - See Figure 7 for location and dimensions of packing groove, setscrew, and mounting bolt circle.

FIGURE 6. Environmental test assembly.

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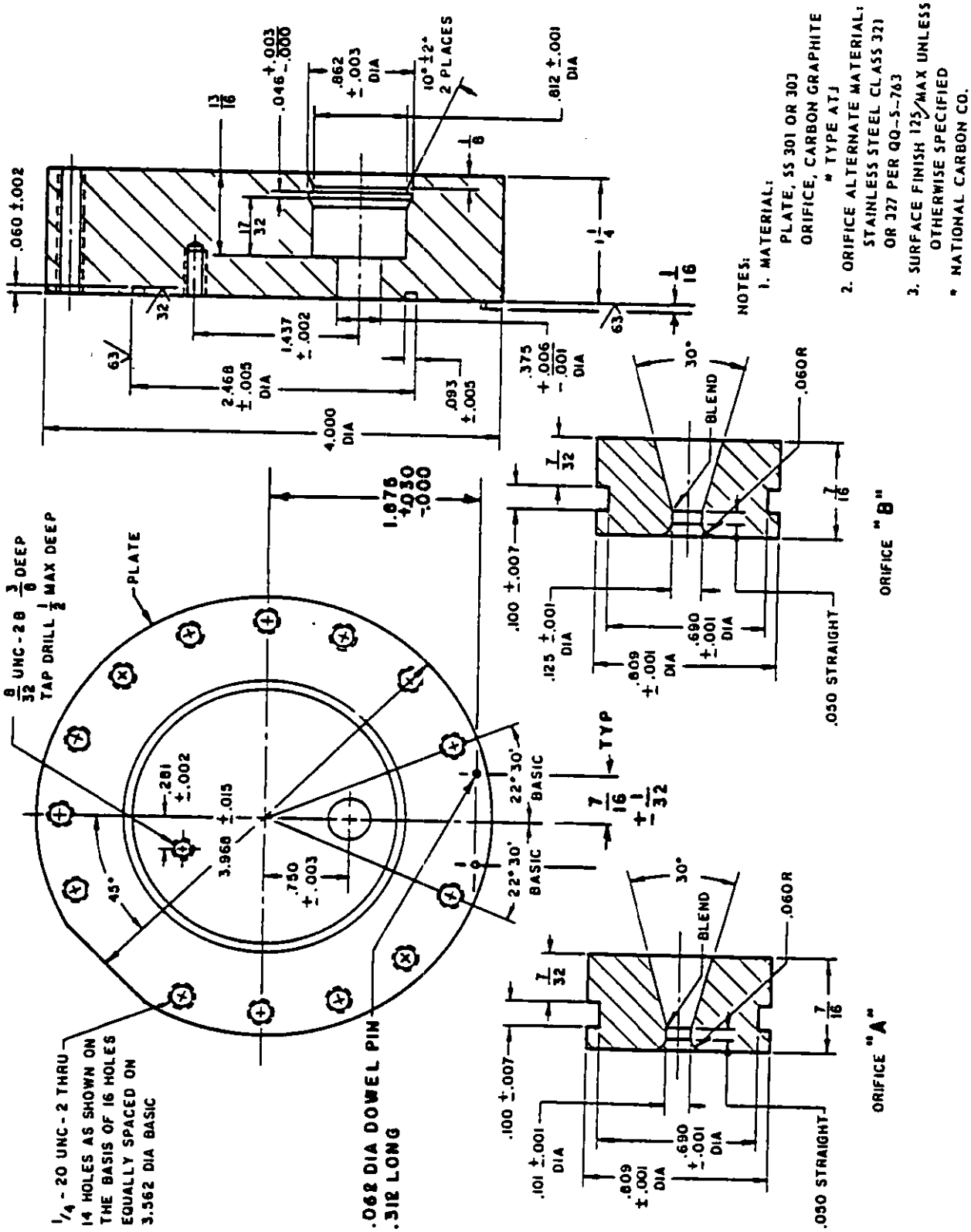


FIGURE 7. Test fixture details.

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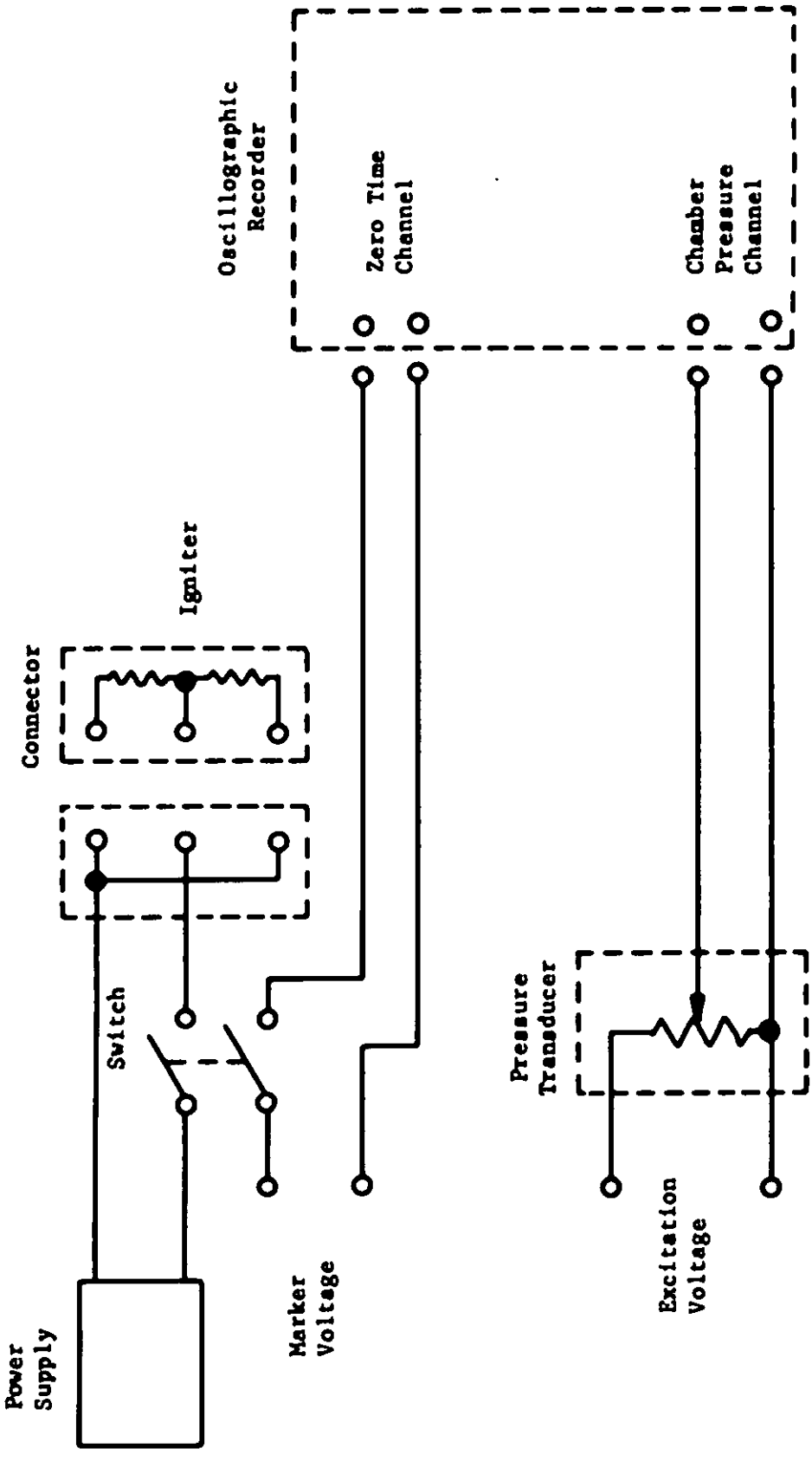


FIGURE 8. Test firing circuit and data recording - analog system option.

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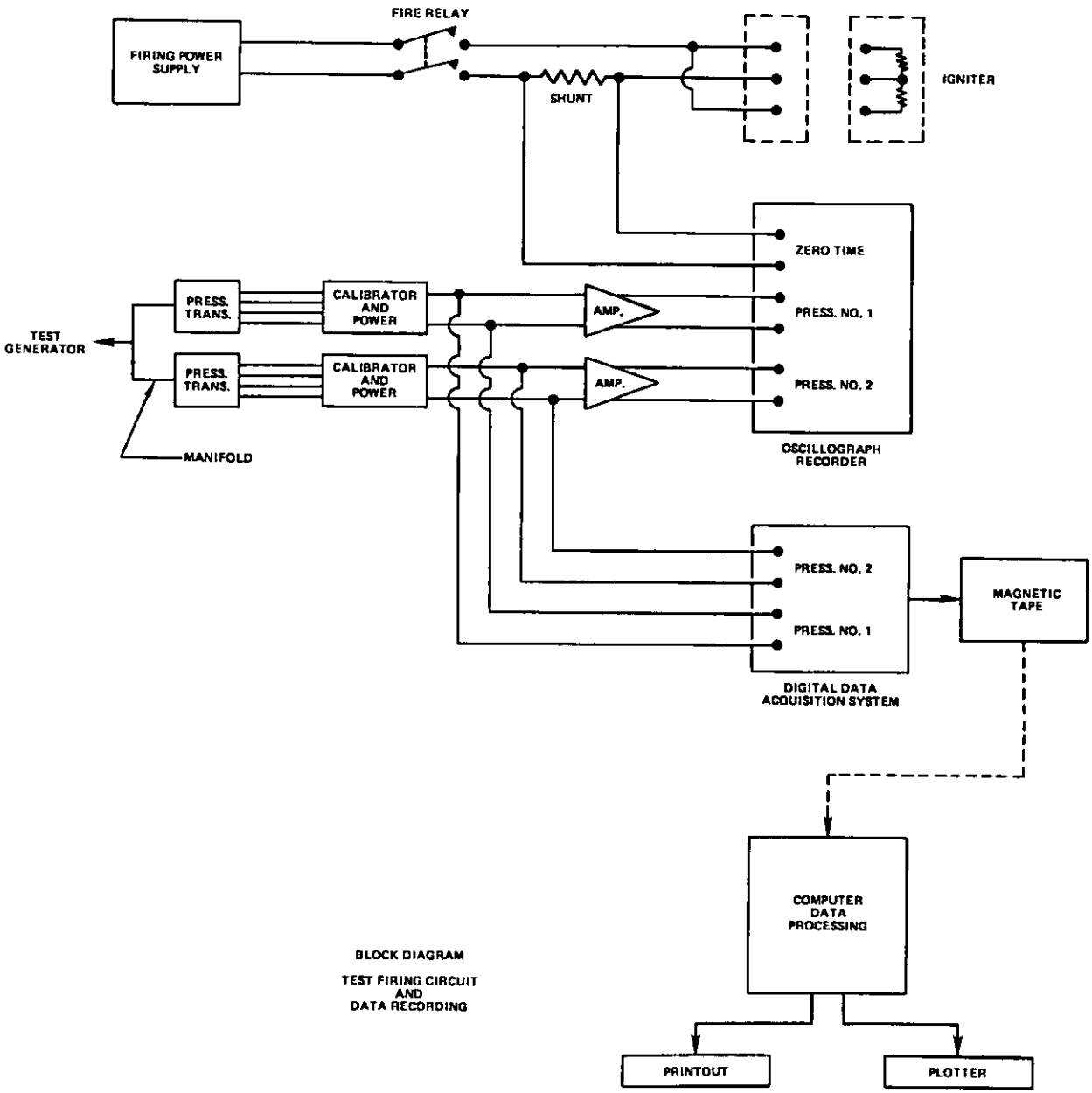


FIGURE 9. Analog and digital system option.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-82438C(OS)	GRAIN, IGNITER, PROPELLANT

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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